WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – I : INTRODUCTION TO HYDROLOGY

1. Engineering Hydrology & Its Applications

> Hydrology is the science that deals with the occurrence, circulation, distribution and properties of water of the earth and its atmosphere. Engineering hydrology = the part concerned with quantity of water for design, planning and operation of water-resource projects (how much, when, where, how often).

❖Applications

- Design of dams, reservoirs and spillways – design flood, reservoir capacity.
- Design of bridges, culverts, drains – waterway area from peak discharge.
- Irrigation planning – crop water requirement, canal capacity.
- Flood control & forecasting, flood-plain zoning, urban drainage.
- Water supply, hydropower, navigation, ground-water development, drought studies.
- Estimation of yield of catchment, sediment, evaporation losses.

❖Branches / terms

- Catchment/ watershed/ basin: area draining to a common outlet. Precipitation, runoff, stream flow, peak flow, base flow.

2. Hydrologic Cycle

> The continuous circulation of water between earth, atmosphere and oceans driven by solar energy and gravity.

- Evaporation (oceans/ lakes/ soil) + transpiration (plants) → water vapour → condensation (clouds) → precipitation → interception, depression storage, infiltration, percolation → ground water, surface runoff → streams → ocean.
- Total earth water ≈ 1.4×10^9 km^3; 97% salt water; 2.5% fresh (mostly ice); ≈ 0.01% usable.

Water balance (hydrologic budget): $P - R - G - E - T = \Delta S$ i.e. Inflow – Outflow = Change in storage

- P = precipitation, R = surface runoff, G = ground-water outflow, E = evaporation, T = transpiration, ΔS = storage change.
- **Horton's engineering representation:** atmospheric, surface, sub-surface phases.

3. Precipitation

Precipitation = all forms of water (liquid or solid) falling from clouds to earth – **rain, snow, hail, sleet, drizzle, glaze, dew/ frost** (the last two are not generally counted).

❖Conditions for precipitation

- Lifting of air (cooling) → saturation → condensation on **nuclei** → drops grow (collision–coalescence, ice crystal process) → fall.

❖Types (by mechanism of lifting)

- **Convective:** warm moist air rises by surface heating, cools, condenses – **thunderstorms**, short, intense, local (tropical).
- **Orographic:** air forced up **mountains** – heavy on windward side (Western Ghats, Himalayas), rain-shadow on lee side.
- **Cyclonic:** low-pressure systems – **frontal** (warm/ cold front) and **non-frontal** (tropical cyclones, monsoon depressions – A.P. & Odisha coast).
- **Monsoon:** seasonal (SW monsoon June–Sept gives ~ 75% of India's rain; NE monsoon Oct–Dec for Tamil Nadu & coastal A.P.).

❖Forms

- **Rain** (> 0.5 mm drops), **drizzle** (< 0.5 mm), **snow**, **sleet** (frozen rain), **hail** (ice balls from cumulonimbus), **glaze**, **dew**, **frost**, **fog drip**.

❖Intensity classification

Class	Rate (mm/h)
Light	< 2.5
Moderate	2.5 – 7.5
Heavy	> 7.5

4. Rainfall Measurement – Rain Gauges

> Rain gauge = instrument measuring rainfall depth (mm) collected over a standard horizontal area (funnel). Depth = volume/ area.

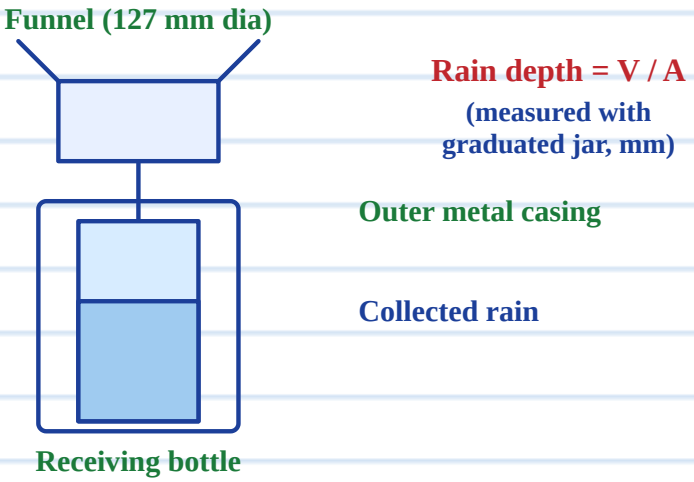


Fig 1.1 Symon's (non-recording) rain gauge

❖A. Non-recording gauges

- **Symon's gauge** (standard in India): **127 mm (5 in.) dia funnel**, bottle, outer casing; read **daily at 08:30 IST** with a graduated measuring jar.
- Limitation: gives only total rainfall; no intensity.

❖B. Recording gauges (give continuous record – hyetograph)

- **Tipping-bucket:** funnel → two small buckets on a pivot; each 0.25/ 0.5 mm fill tips & records electrically. Good for telemetry.
- **Weighing-bucket:** weight of collected rain/ snow recorded by pen on a drum; best for snow.
- **Float (natural syphon / Dines tilting-siphon):** float rises with water, pen traces mass curve on a clock-driven chart; when full the **siphon** empties it automatically – used by IMD.
- **Telemetering / automatic weather stations (AWS)**, radar & satellite rainfall estimates.

❖Siting of a rain gauge (important)

- Open, level ground, away from trees/ buildings – **distance ≥ 2 × height of obstruction**; funnel rim horizontal, **30 cm above ground** (to avoid splash).
- **Density of gauges (IS 4987 / WMO):** plains 1 per 520 km^2; hilly 1 per 260 km^2; heavy-rain hills 1 per 130 km^2; at least 10% recording.
- Wind-shield, error due to wind eddies, evaporation, splash, wetting loss.

5. Presentation & Interpretation of Rainfall Data

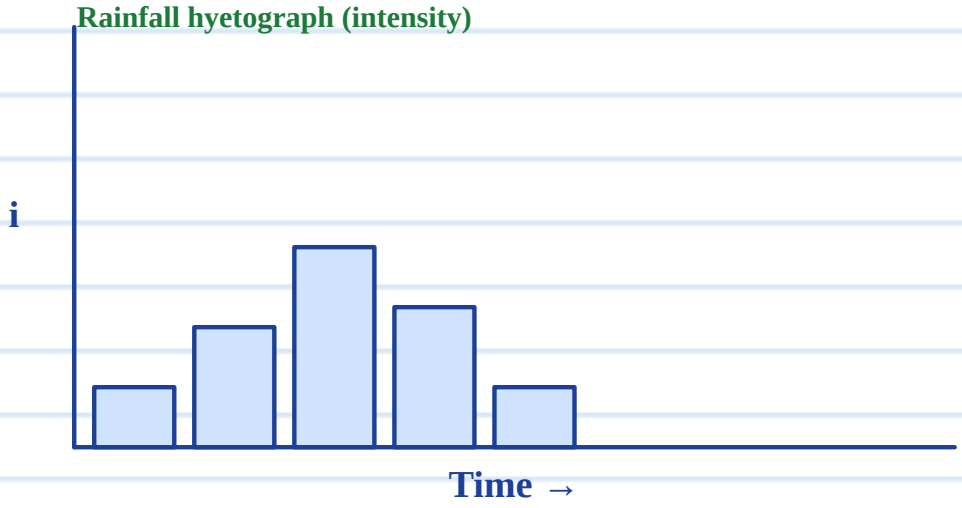


Fig 1.3 Hyetograph – rainfall intensity versus time

- **Hyetograph:** bar graph of rainfall intensity (mm/h) vs time – shows storm pattern.
- **Mass curve of rainfall:** cumulative rainfall vs time – slope = intensity; from recording gauge.
- **Point rainfall** table; **Isohyets** (lines of equal rainfall) maps.
- **Depth–Area–Duration (DAD) curves:** average depth reduces as area increases; used for design storms.
- **Intensity–Duration–Frequency (IDF) curves:** $i = K T^x/(t + a)^n$.
- **Recurrence interval (return period)** $T = (N+1)/m$ (**Weibull**; m = rank in descending order); probability of exceedance $P = 1/T$; risk of occurrence at least once in n years = $1 - (1 - 1/T)^n$.

❖Missing rainfall data

- Station X record missing; nearby stations A, B, C with normal annual rainfalls N_A, N_B, N_C and storm rainfalls P_A, P_B, P_C:

$$P_X = (N_X/n)\,[P_A/N_A + P_B/N_B + P_C/N_C]$$ (if normals differ > 10%: normal-ratio method; else simple arithmetic mean)

❖Consistency of record – double mass curve

- Plot **cumulative annual rainfall of station X vs cumulative average of surrounding stations**; change in **slope** indicates change in gauge conditions/ location. Correct: $P_c = P_a \times (M_c/M_a)$ (slopes).
- **Test of adequacy of gauges:** $N = (C_v/\varepsilon)^2$; $C_v = 100\,\sigma/\bar{P}$; ε = permissible error (10%).

6. Average Rainfall over a Basin

❖(a) Arithmetic mean method

$$\bar{P} = (P_1 + P_2 + \dots + P_n) / n$$

- Simple; suitable for flat area with uniformly distributed gauges & small variation (< 10%).

❖(b) Thiessen polygon method

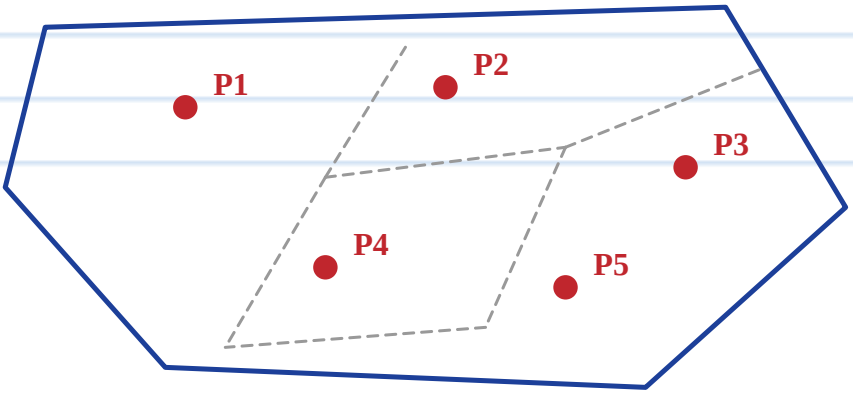


Fig 1.2 Thiessen polygon method (area-weighted rainfall)

- Join adjacent gauges; draw **perpendicular bisectors** → polygons; area A_i of each polygon around gauge i.

$$\bar{P} = \Sigma (P_i A_i) / \Sigma A_i$$

- Accounts for non-uniform gauge distribution; **more accurate than arithmetic**; doesn't consider orography; polygons change if a gauge is added/ missing.

❖(c) Isohyetal method

- Draw **isohyets** (equal rainfall contours) by interpolation; area A_j between successive isohyets; mean rainfall of each strip $P_j = (P_a + P_b)/2$.

$$\bar{P} = \Sigma (P_j A_j) / \Sigma A_j$$

- **Most accurate** (esp. in hills with orographic effects) but needs judgement and many gauges.

❖Comparison

Method	Suitable for	Accuracy
Arithmetic	Flat, uniform gauges	Low
Thiessen	Moderately varied terrain	Medium
Isohyetal	Hilly/ large area	Highest

❖Example

> Four gauges with rainfall (mm) & Thiessen areas (km^2): 80/ 200, 100/ 300, 120/ 400, 60/ 100. $\Sigma P A = 16000 + 30000 + 48000 + 6000 = 100{,}000$; $\Sigma A = 1000 \rightarrow \bar{P} = 100$ mm. Arithmetic mean = (80+100+120+60)/4 = 90 mm.

7. Evaporation

> **Evaporation** = the process by which **liquid water changes to vapour** at a free surface (water, wet soil) below boiling point; needs **latent heat** (≈ 2.45 MJ/kg at 20 °C). **Evapotranspiration (ET)** = evaporation + plant transpiration.

- **Importance:** reservoir losses (India: 1–3 m/yr from reservoirs!), irrigation scheduling, water balance.

❖Factors affecting evaporation

- **Vapour pressure difference** ($e_w - e_a$): e_w = saturated vapour pressure at water surface temperature; e_a = actual vapour pressure of air – **driving force** (Dalton's law).
- **Temperature** (air & water) – higher $T \rightarrow$ higher $e_w \rightarrow$ more evaporation.
- **Wind speed** – removes saturated air layer; E increases with wind up to a limit.
- **Relative humidity** – lower RH, more evaporation.
- **Atmospheric pressure** – lower pressure (altitude) $\rightarrow$ more evaporation.
- **Quality of water** – dissolved salts reduce vapour pressure (sea water ≈ 2–3% less than fresh).
- **Size & depth of water body**, solar radiation, heat storage; **monolayers** (cetyl alcohol) reduce loss.

❖Equations

Dalton: $E = C\,(e_w - e_a)$ (E in mm/day, e in mm Hg)

Meyer's formula: $E_L = K_M\,(e_w - e_a)\,(1 + u_9/16)$ mm/day

- u_9 = wind speed (km/h) at 9 m above ground; K_M = 0.36 (large deep waters), 0.50 (small shallow waters).
- Other methods: water-budget, energy-budget, Penman, mass-transfer, empirical (Rohwer).

❖Measurement of evaporation

- Evaporation pans:
  - Class A land pan (US Weather Bureau): 1210 mm dia × 255 mm deep, galvanised iron, on wooden platform; water depth 18–20 cm; hook gauge reading.
  - ISI (Indian Standard) pan: 1220 mm dia × 255 mm, with a fine-wire mesh cover; pan coefficient 0.8.
  - Colorado sunken pan, floating pan (best for lakes).
- Lake evaporation = pan coefficient × pan evaporation: $E_L = C_p\,E_p$; $C_p \approx 0.70$ (Class A), 0.8 (ISI), 0.78 (Colorado), 0.80 (floating).
- Atmometers/ evaporimeters (Piche) for lab; lysimeters for ET.
- Control of evaporation: reduce exposed area (deep narrow reservoirs), wind-breaks, monomolecular films (hexadecanol), storing water underground, spray shading.

❖Example

> Class A pan evaporation = 8 mm/day, C_p = 0.7 → lake evaporation = 5.6 mm/day.

> Meyer: e_w = 20 mm Hg, e_a = 10 mm Hg, u_9 = 16 km/h, K_M = 0.36: E = 0.36 × 10 × (1 + 1) = 7.2 mm/day.

8. Infiltration

> Infiltration = movement of water from the ground surface into the soil. Infiltration rate f (mm/h) = depth entering per unit time; infiltration capacity f_p = maximum rate under given conditions. Percolation = downward movement within soil to water table.

❖Factors affecting infiltration

- Soil type & texture (sand high, clay low), soil structure/ porosity, initial moisture content (dry soil – high initial rate).
- Surface cover/ vegetation (roots & litter ↑), land use, compaction/ crusting (rain-drop sealing ↓), slope.
- Rainfall intensity (rate cannot exceed supply), depth of ponding, temperature/ viscosity, entrapped air, quality of water (silt, salts), tillage.

❖Horton's infiltration equation

$$f_p = f_c + (f_0 - f_c)\, e^{-kt}$$

- f_0 = initial capacity, f_c = final (constant) capacity, k = decay constant (h^{-1}); rate decreases exponentially with time as soil fills.
- Cumulative infiltration $F(t) = f_c\, t + (f_0 - f_c)(1 - e^{-kt})/k$. Other models: Green–Ampt, Philip, Kostiakov.

❖Measurement of infiltration

- Double-ring infiltrometer: two concentric rings (30 cm inner, 60 cm outer) driven in soil; water maintained at constant depth; outer ring prevents lateral spreading; inner ring reading of water added at intervals gives f.
- Single-ring infiltrometer (lateral spread error); rainfall simulators (sprinkler plots); hydrograph analysis; tension infiltrometer.

❖Infiltration indices (avg losses over a storm)

- Φ-index: the constant rate of rainfall above which the volume of rainfall = volume of direct runoff. Rain below Φ is assumed all infiltration. Find by trial: $\Sigma\ (i - \Phi)\,\Delta t = R$ (runoff depth) for intervals with $i > \Phi$.
- W-index: average infiltration rate = $(P - R - S_d)/t_r$, corrected for depression storage S_d and rainfall duration where $i > f$. $W > \Phi$ (since W removes initial losses).

- **Φ-min index** for wet conditions.

❖Example (Φ-index)

> Hourly rainfall (cm): 1.0, 2.0, 3.0, 2.0, 1.0; runoff depth $R = 4.0$ cm. Assume $\Phi = 1.5$: $\Sigma(i - 1.5)$ for $i > \Phi = (2-1.5)+(3-1.5)+(2-1.5) = 2.5$ (but 1.0 below) ✗. Try $\Phi = 1.0$: $(1+2+1) = 4.0$ ✔ ($i = 1.0 \rightarrow 0$). Hence $\Phi = 1.0$ cm/h.

9. Runoff

> **Runoff** = that part of precipitation that appears in streams/ rivers as surface flow. **Total runoff** = surface runoff + interflow + base flow (ground-water flow). **Direct runoff** = surface + prompt interflow.

❖Factors affecting runoff

- **Climatic:** rainfall **intensity, duration, areal distribution,** storm direction, antecedent rainfall, evaporation, temperature, season.
- **Catchment (physiographic): area, shape** (fan-shaped gives high peak vs elongated), **slope, elevation, orientation, soil type & geology,** land use & vegetation, **drainage density,** depression storage, lakes/ ponds, urbanisation (impervious cover ↑ runoff).
- **Hydraulic:** channel storage, stream density, infiltration.

❖Computation of runoff (volume / yield)

- **Strange's table** (rainfall, soil class, yield %), **Binnie's percentages, Barlow's coefficients** (runoff coefficient from table; $Q = K_b P$), **Khosla's formula** (temperature based):

> Khosla's formula: $R = P - L$ (L = evaporation/ infiltration loss, estimated from mean temperature T of the catchment)

- **Inglis & De Souza formula (Maharashtra):** $R = 0.85P - 30.5$ (Ghat regions) ; $R = (1/254)\,P\,(P - 17.8)$ (plains) (R, P in cm).
- **SCS-CN (Curve Number) method (USDA):**

> $Q = (P - I_a)^2 / (P - I_a + S)$, $I_a = 0.2S \rightarrow Q = (P - 0.2S)^2 / (P + 0.8S)$; $S = 25400/CN - 254$ (mm)

- CN = 100 (impervious) ... 30; depends on soil group A–D, land use, antecedent moisture condition (AMC I, II, III); valid when $P > 0.2S$.

❖Example (SCS)

> $P = 100$ mm, $CN = 80$: $S = 25400/80 - 254 = 63.5$ mm; $I_a = 12.7$; $Q = (100 - 12.7)^2/(100 + 50.8) = 7621/150.8 = 50.5$ mm (runoff coefficient ≈ 0.5).

❖Runoff coefficient

C = Runoff / Rainfall ; typical C: roofs 0.9; pavements 0.8; sandy lawns 0.1–0.2; clay 0.5; forests 0.1–0.2; cultivated 0.3

10. Design Flood & Maximum Rate of Run-off

> Design flood = the flood adopted for the design of a hydraulic structure (spillway, bridge, weir) corresponding to a safe, economical risk (return period). Types: Probable Maximum Flood (PMF), Standard Project Flood (SPF), Maximum Probable Flood (MPF); frequency flood (e.g., 100-yr, 1000-yr).

- Return period (IS 11223): large dams (PMF), small structures (25–100 yr), bridges (50–100 yr), culverts (25 yr), urban drains (2–10 yr).

❖Estimation of maximum rate of runoff (peak discharge)

(a) Empirical formulas (Q_{max} in m^3/s, A in km^2)

- Dicken's: $Q = C_D A^{3/4}$ (C_D = 11.4 for N. India hills, 13.9–19.5 central, 22–25 W. Ghats, A in km^2).
- Ryves: $Q = C_R A^{2/3}$ (C_R = 6.8 near coast (80 km), 8.5 (80–160 km), 10.1 (limited hills)).
- Inglis: $Q = 124 A / \sqrt{(A + 10.4)}$ (Western Ghats – Maharashtra).
- Nawab Jung Bahadur (Hyderabad/ A.P.): $Q = C_N A^{(0.993 - 1/14 \log A)}$, C_N = 48 – 60.
- Limitations: region-specific, ignore storm characteristics.

(b) Rational method (small catchments < 50 km^2)

$Q_p = C i A / 360$ (Q in m^3/s, i in mm/h, A in hectares) ≡ $Q = 0.278 C i A$ (A in km^2)

- i = rainfall intensity for duration = time of concentration t_c (time for water from the most remote point to reach outlet). Kirpich: $t_c = 0.0195\,L^{0.77}\,S^{-0.385}$ (min; L in m, S slope m/m).
- Assumptions: uniform intensity over whole area for duration t_c, C constant.

(c) Unit hydrograph method, (d) Flood-frequency analysis (Gumbel, log-Pearson III), (e) Envelope curves.

❖Gumbel's method (flood frequency)

$$x_T = \bar{x} + K\sigma\,,\quad K = -(\sqrt{6}/\pi)\,[\,0.5772 + \ln\ln(\,T/(T-1)\,)\,]$$

❖Examples

Dicken's: $A = 100\ km^2$, $C_D = 14 \rightarrow Q = 14 \times 100^{0.75} = 14 \times 31.6 = 443\ m^3/s$.

Rational: $A = 200$ ha, $C = 0.5$, $i = 36$ mm/h $\rightarrow Q = 0.5\times36\times200/360 = 10\ m^3/s$.

Inglis: $A = 50\ km^2 \rightarrow Q = 124\times50/\sqrt{60.4} = 798\ m^3/s$.

II. Separation of Base Flow

Hydrograph separation = splitting the stream flow hydrograph into direct runoff (DRH) and base flow (ground-water contribution); needed to derive unit hydrographs.

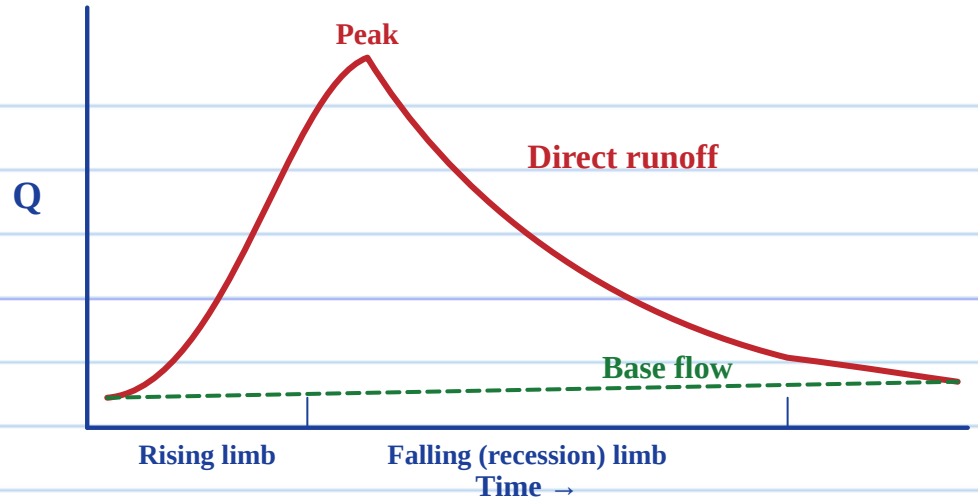


Fig 1.4 Hydrograph with base-flow separation (straight-line method)

❖Methods

- **Straight-line method (Method I):** join the start of rising limb (A) to the point on recession limb (B) where direct runoff ends, by a horizontal/ straight line; area above = DRH. Simple.
- **Fixed base length:** end of direct runoff after N days = $0.827\,A^{0.2}$ (A in km²); line from A (start) extended; the recession curve extended backward from point of inflection.
- **Variable slope method:** project the pre-storm recession curve to the peak time, join to the point N days after the peak.
- **Recession (master) curve analysis:** $Q_t = Q_0\,K_r^t$ (K_r = recession constant 0.93–0.95 for storage); plot **log Q vs t**; break in slope shows end of surface runoff.
- **Direct-runoff depth** = volume of DRH / catchment area.

$$\text{Direct runoff depth (cm)} = [\Sigma\ \text{DRH ordinates } (m^3/s) \times \Delta t\ (s)] / [A\ (m^2)] \times 100$$

❖Example

Catchment 36 km². DRH ordinates sum 90 m³/s at 6-h intervals → volume = $90\times6\times3600 = 1.944\times10^6$ m³. Depth = $1.944\times10^6/36\times10^6 = 0.054$ m = **5.4 cm.**

12. Important Questions – Unit I

- Define hydrology; applications. Explain the hydrologic cycle with a sketch.
- Types and forms of precipitation. Describe recording & non-recording rain gauges.
- Methods of computing average rainfall (Thiessen, isohyetal); double mass curve; missing data.
- Factors affecting evaporation; measurement; Meyer's formula; control.
- Infiltration – factors, measurement (double ring), Horton equation, Φ & W indices.
- Factors affecting runoff; SCS-CN; empirical formulas; rational method.
- Design flood; separation of base flow.

Formula sheet: $\bar{P} = \Sigma PA/\Sigma A$; $Q = CiA/360$; $S = 25400/CN - 254$; $f = fc + (f0 - fc)e^{(-kt)}$; $N = 0.827A^{0.2}$